1 Mathematical model and hybrid mechanistic–neural formulation

The climate-driven mosquito SEI–human SEIR transmission structure is adapted from [1], whereas the explicit mechanistic mosquito recruitment term is replaced here by a constrained neural representation of effective adult recruitment. The prescribed thermal-response functions are taken from experimentally fitted relationships. This separation retains experimentally supported biological mechanisms while allowing the comparatively difficult-to-observe climate-to-recruitment relationship to be learned from data, following the general framework of universal differential equations and recent mechanistic–neural mosquito models [2, 3].

The study considers five Brazilian cities, whose identifiers and modelling roles are summarized in Table 2. The same model formulation is applied separately to each city. To simplify the notation, the city index is omitted in the present section; therefore, N_h, the meteorological inputs, the initial conditions, the recruitment scale, and all model trajectories should be understood as city-specific. No movement of humans or mosquitoes between cities is represented.

The adult female mosquito population follows an SEI structure, whereas the human population follows an SEIR structure, consistently with established dengue transmission models [4, 5, 6, 1]. Only adult female mosquitoes are represented because blood-feeding females acquire and transmit dengue virus [7, 1].

Calendar time t is measured in *days*. An epidemiological week is the interval $[t_w, t_{w+1})$ with $t_{w+1} - t_w = 7$ days. Consequently, every rate appearing in the differential system is expressed per day, while reported cases and meteorological covariates are indexed by epidemiological week.

The state vector is

$$\boldsymbol{x}(t) = \left(S_m(t), E_m(t), I_m(t), S_h(t), E_h(t), I_h(t), R_h(t)\right)^\top, \tag{1}$$

where S_m, E_m, and I_m denote susceptible, exposed, and infectious adult female mosquitoes, respectively. The variables S_h, E_h, I_h, and R_h denote susceptible, exposed, infectious, and recovered humans. The total populations are

$$N_m(t) = S_m(t) + E_m(t) + I_m(t), \qquad N_h = S_h(t) + E_h(t) + I_h(t) + R_h(t). \tag{2}$$

Biological assumptions. The primary model relies on the following assumptions.

1. **Balanced human demography.** Human recruitment balances natural mortality at rate $\mu_h N_h$, so N_h remains constant. This closure is standard in deterministic dengue models [5].

2. **Effective single-serotype dynamics.** The four dengue serotypes are not modelled separately. The recovered class represents effective immunity over the analysed horizon, as in single-serotype SEIR formulations [6]. Changes in the dominant serotype, cross-immunity, antibody-dependent enhancement, and secondary-infection effects are therefore outside the present model and constitute limitations [8].

3. **Homogeneous within-city mixing.** Humans and mosquitoes are assumed to mix homogeneously at the city scale, yielding frequency-dependent forces of infection [5, 1]. The exclusion of between-city movement is a spatial closure adopted in the present study.

4. **No vertical transmission.** Newly recruited adult females enter the susceptible mosquito compartment. Vertical dengue-virus transmission is omitted because the low efficiencies commonly reported are not expected to dominate long-term viral persistence [9].

5. **Exogenous climate.** Temperature, precipitation, and relative humidity are externally observed forcings and are not affected by the epidemic state [6, 1].

The compartmental transitions, the two directions of dengue transmission, and the neural climate-to-recruitment component are summarized in Figure 1.

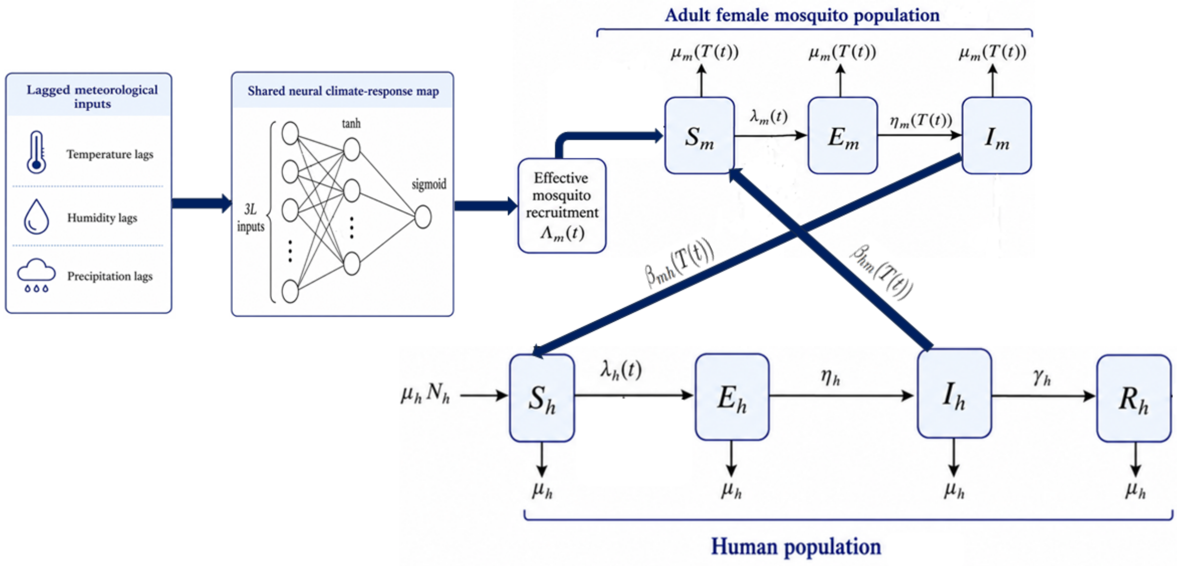

Figure 1: Hybrid mechanistic–neural structure of the coupled adult-mosquito SEI and human SEIR dengue transmission model. The diagram shows the compartmental transitions, human-to-mosquito and mosquito-to-human transmission, and the neural mapping from lagged meteorological covariates to effective adult mosquito recruitment.

For $t \in [t_w, t_{w+1})$, where w denotes the epidemiological week, the complete model for a generic city is

$$\begin{cases} \dfrac{\mathrm{d}S_m}{\mathrm{d}t} = \Lambda_m(t) - a(T(t))\,\beta_{hm}(T(t))\,\dfrac{I_h(t)}{N_h} S_m(t) - \mu_m(T(t))\,S_m(t), \\ \dfrac{\mathrm{d}E_m}{\mathrm{d}t} = a(T(t))\,\beta_{hm}(T(t))\,\dfrac{I_h(t)}{N_h} S_m(t) - [\eta_m(T(t)) + \mu_m(T(t))]\,E_m(t), \\ \dfrac{\mathrm{d}I_m}{\mathrm{d}t} = \eta_m(T(t))\,E_m(t) - \mu_m(T(t))\,I_m(t), \\ \dfrac{\mathrm{d}S_h}{\mathrm{d}t} = \mu_h N_h - a(T(t))\,\beta_{mh}(T(t))\,\dfrac{I_m(t)}{N_h} S_h(t) - \mu_h S_h(t), \\ \dfrac{\mathrm{d}E_h}{\mathrm{d}t} = a(T(t))\,\beta_{mh}(T(t))\,\dfrac{I_m(t)}{N_h} S_h(t) - (\eta_h + \mu_h)\,E_h(t), \\ \dfrac{\mathrm{d}I_h}{\mathrm{d}t} = \eta_h E_h(t) - (\gamma_h + \mu_h)\,I_h(t), \\ \dfrac{\mathrm{d}R_h}{\mathrm{d}t} = \gamma_h I_h(t) - \mu_h R_h(t). \end{cases} \tag{3}$$

The mosquito infection term describes acquisition of dengue virus by a susceptible mosquito feeding on an infectious human. Here, $a(T(t))$ is the number of blood meals taken by one adult female per day, $\beta_{hm}(T(t))$ is the probability of human-to-mosquito infection per infectious blood meal, and $I_h(t)/N_h$ is the infectious human prevalence. Conversely, the human infection term describes exposure of susceptible humans to infectious mosquitoes. The quantity $\beta_{mh}(T(t))$ is the probability of mosquito-to-human transmission per bite. The subscripts indicate the direction of transmission: $h \to m$ for human-to-mosquito transmission and $m \to h$ for mosquito-to-human transmission [7, 6].

Exposed humans progress to infectiousness at rate η_h, and infectious humans recover at rate γ_h. Balanced recruitment and natural mortality imply

$$\frac{\mathrm{d}N_h}{\mathrm{d}t} = 0. \tag{4}$$

The time-independent human parameters are reported in Table 1. The mortality rate is approximated by the reciprocal of the Brazilian life expectancy reported for 2022. This constant-hazard representation is a modelling approximation, whereas the life-expectancy value is taken from the official complete life tables [10].

Table 1: Time-independent human parameters. All rates are expressed per day.

Symbol	Definition	Primary value or expression	Unit	Reference
N_h	Constant human population of the city	Municipality population in 2010 from the demographic table	persons	[11]
η_h	Progression rate from exposed to infectious	$\eta_h = \dfrac{1}{5.9} \simeq 0.1695$	day^{-1}	[12, 6]
γ_h	Human recovery rate	$\gamma_h = \dfrac{1}{5.0} = 0.2000$	day^{-1}	[6]
μ_h	Human natural mortality rate	$\mu_h = \dfrac{1}{75.5 \times 365} \simeq 3.6288 \times 10^{-5}$	day^{-1}	[10]

The quantities $a(T)$, $\beta_{mh}(T)$, $\beta_{hm}(T)$, $\eta_m(T)$, and $\mu_m(T)$ are fixed biological functions of temperature. Their functional parameters are not re-estimated from the dengue case series. In contrast, the recruitment term depends on the weekly meteorological vector $\boldsymbol{z}_w$ and on trainable neural parameters $\boldsymbol{\theta}$.

1.1 Climate-driven effective mosquito recruitment

The influx of newly recruited adult female mosquitoes is defined by

$$\Lambda_m(t) = \kappa N_h q_{\boldsymbol{\theta}}(\boldsymbol{z}_w), \qquad t \in [t_w, t_{w+1}), \tag{5}$$

where $\kappa > 0$ is a city-specific recruitment scale, with units mosquitoes person^{-1} day^{-1}, and $q_{\boldsymbol{\theta}} : \mathbb{R}^{3L} \to (0,1)$ is a climate-response function shared across the development cities. The city index is omitted in Equation (5); κ_c is estimated separately for each city, whereas $\boldsymbol{\theta}$ is common to the four development cities.

The neural map is

$$q_{\boldsymbol{\theta}}(\boldsymbol{z}) = \sigma\left(\boldsymbol{w}_2^\top \tanh(W_1 \boldsymbol{z} + \boldsymbol{b}_1) + b_2\right), \qquad \sigma(u) = \frac{1}{1+\exp(-u)}, \tag{6}$$

where $\boldsymbol{\theta} = \{W_1, \boldsymbol{b}_1, \boldsymbol{w}_2, b_2\}$ contains the trainable weights and biases. The bounded sigmoid output follows the strategy of embedding a constrained neural representation of an incompletely specified process within a mechanistic system [2, 3]. It ensures

$$0 < q_{\boldsymbol{\theta}}(\boldsymbol{z}) < 1, \qquad 0 < \Lambda_m(t) < \kappa N_h. \tag{7}$$

For a memory window of L epidemiological weeks, the input vector is

$$\boldsymbol{z}_w = \left[\widetilde{T}_w, \ldots, \widetilde{T}_{w-L+1}, \widetilde{P}^\star_w, \ldots, \widetilde{P}^\star_{w-L+1}, \widetilde{H}_w, \ldots, \widetilde{H}_{w-L+1}\right]^\top \in \mathbb{R}^{3L}, \tag{8}$$

where T_w is weekly mean temperature, P_w is weekly accumulated precipitation, and H_w is weekly mean relative humidity. Relative humidity is retained instead of saturation vapour-pressure deficit because it is directly provided in the harmonized dataset and avoids an additional derived transformation. Lagged meteorological information is included because climate may influence mosquito reproduction, immature-stage survival, adult emergence, and dengue transmission after delays [1, 3].

Because precipitation is non-negative and right-skewed, it is transformed as

$$P_w^\star = \log\left(1 + \frac{P_w}{P_0}\right), \qquad P_0 = 1 \text{ mm}, \tag{9}$$

where P_0 makes the logarithm argument dimensionless. This shifted logarithmic transformation is a study-specific variance-compression choice consistent with standard transformations of skewed positive variables [13].

Each meteorological variable is standardized using pooled observations from the four development cities during the initial training period 2011–2018:

$$\widetilde{X}_{c,w} = \frac{X_{c,w} - \overline{X}_{\text{dev,train}}}{s_{X,\text{dev,train}}}, \qquad X \in \{T, P^\star, H\}, \tag{10}$$

where $\overline{X}_{\text{dev,train}}$ and $s_{X,\text{dev,train}}$ are the pooled mean and standard deviation. These statistics are fixed after their calculation and are applied unchanged to the validation period, the final refitting period, and Brasília. This prevents validation or test information from entering preprocessing [14, 15].

A single-hidden-layer feedforward network is used because such networks can approximate broad classes of continuous functions [16]. The lag window and hidden-layer width are not treated as biological constants. Candidate windows $L \in \{3, 5, 7\}$ and candidate hidden widths $J \in \{4, 8\}$ are compared on the 2019–2020 validation period. The selected architecture is reported in the Results section and is fixed before the final test.

The term *effective recruitment* is essential. In the absence of continuous entomological observations in every city, $\Lambda_m(t)$ is not interpreted as the literal number of newly emerged females. It represents the net adult recruitment pressure compatible with the observed dengue counts, lagged meteorological inputs, prescribed thermal responses, observation process, and model assumptions. It may therefore absorb unresolved effects of aquatic habitat availability, immature-stage survival, female emergence, host accessibility, mosquito-control activities, and within-city spatial heterogeneity. Restricting the neural component to this single latent mechanism preserves the interpretation of the prescribed biting, transmission, incubation, and mortality processes and limits compensation among several simultaneously learned biological mechanisms [17].

1.2 Fixed thermal-response functions

The thermal responses are fixed at published experimental estimates rather than inferred from the epidemiological observations. This preserves their biological interpretation and reduces the number of competing climate-dependent mechanisms estimated from the same incidence series [17].

For biting, transmission, and extrinsic incubation, the asymmetric Brière response is used [18]:

$$\mathcal{B}(T; c, T_{\min}, T_{\max}) = \begin{cases} cT(T - T_{\min})\sqrt{T_{\max} - T}, & T_{\min} < T < T_{\max}, \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

The projections

$$[x]_+ = \max\{0, x\}, \qquad [x]_0^1 = \min\{1, \max\{0, x\}\} \tag{12}$$

ensure non-negative rates and constrain per-bite probabilities to $[0, 1]$. The following values are the posterior-mean thermal-response parameters reported for *Aedes aegypti* and dengue virus by Mordecai et al. [7, 19] and used subsequently in a dynamic SEI–SEIR model [6].

Biting rate.

$$a(T) = \left[\mathcal{B}\left(T; 2.02 \times 10^{-4}, 13.35, 40.08\right)\right]_+, \tag{13}$$

where $a(T)$ has units bites mosquito^{-1} day^{-1} [7, 6].

Mosquito-to-human transmission probability.

$$\beta_{mh}(T) = \left[\mathcal{B}\left(T; 8.49 \times 10^{-4}, 17.05, 35.83\right)\right]_0^1. \tag{14}$$

This quantity corresponds to the thermal trait denoted by $b(T)$ in the source transmission models [7, 6].

Human-to-mosquito transmission probability.

$$\beta_{hm}(T) = \left[\mathcal{B}\left(T; 4.91 \times 10^{-4}, 12.22, 37.46\right)\right]_0^1. \tag{15}$$

This quantity corresponds to the probability of mosquito infection, $\mathrm{pMI}(T)$, in the source models [7, 6].

Extrinsic incubation rate.

$$\eta_m(T) = \left[\mathcal{B}\left(T; 6.65 \times 10^{-5}, 10.68, 45.90\right)\right]_+, \tag{16}$$

where $\eta_m(T)$ has units day^{-1} [7, 6]. Under the exponential exposed-stage assumption, the mean extrinsic incubation period is $1/\eta_m(T)$ whenever $\eta_m(T) > 0$. The temperature dependence is consistent with experimental evidence that dengue-virus extrinsic incubation shortens at biologically suitable higher temperatures [12].

Adult female mosquito mortality. Adult mosquito mortality is represented by the fourth-degree polynomial fitted by Yang et al. to temperature-controlled observations of adult female *Aedes aegypti* [20]:

$$p_\mu(T) = 0.8692 - 0.1590T + 0.01116T^2 - 3.408 \times 10^{-4}T^3 + 3.809 \times 10^{-6}T^4. \tag{17}$$

Atmospheric moisture can affect adult survival through desiccation stress [21]; however, the primary mortality function is kept temperature-only because it is directly calibrated to species- and sex-specific controlled experiments and avoids estimating an additional survival mechanism from the same incidence data [20, 1].

To prevent unsupported polynomial extrapolation, temperature is projected onto the experimental interval:

$$\Pi^\mu(T) = \min\left\{T^\mu_{\max}, \max\left\{T^\mu_{\min}, T\right\}\right\}, \tag{18}$$

where $T^\mu_{\min} = 10.54\,^\circ\mathrm{C}$ and $T^\mu_{\max} = 33.41\,^\circ\mathrm{C}$ correspond to the temperature range represented in the adult-female experiments [20]. The mortality rate used in the dynamic system is

$$\mu_m(T) = p_\mu(\Pi^\mu(T)). \tag{19}$$

The projection is a numerical modelling choice introduced in the present study. It leaves the fitted polynomial unchanged inside its calibration interval while preventing unsupported extrapolation outside that interval.

1.3 Observation model for weekly probable dengue cases

The surveillance data consist of probable dengue cases aggregated by municipality and epidemiological week of symptom onset [11, 22]. The city index c is reintroduced because the observed counts and observation parameters are city-specific.

The ODE system describes latent infections. Since η_h represents progression associated with the intrinsic incubation period, the weekly number of individuals progressing from the exposed to the infectious class is used as a proxy for symptom onset:

$$C_{c,w} = \int_{t_w}^{t_{w+1}} \eta_h E_{h,c}(t)\,\mathrm{d}t. \tag{20}$$

This alignment is consistent with the definition of the intrinsic incubation period [12] and with mechanistic dengue observation models that relate reported cases to transitions from the exposed to the infectious class

[23]. At weekly resolution, it remains an approximation because human infectiousness to mosquitoes may begin shortly before symptom onset [24].

Let $Y_{c,w}$ denote the observed number of probable dengue cases in city c during week w. The latent incidence is linked to surveillance counts through

$$Y_{c,w} \sim \text{NegBin}\left(m_{c,w}, \phi_c\right), \qquad m_{c,w} = \rho_c C_{c,w}, \tag{21}$$

using the mean–dispersion parameterization

$$\mathbb{E}[Y_{c,w}] = m_{c,w}, \qquad \text{Var}(Y_{c,w}) = m_{c,w} + \frac{m_{c,w}^2}{\phi_c}. \tag{22}$$

The effective observation probability $\rho_c \in (0,1)$ represents the combined probability that an infection produces clinically observable symptoms, leads to healthcare attendance, is classified as a probable dengue case, and is recorded by the surveillance system. Passive dengue surveillance is known to be incomplete and its accuracy may vary by setting and epidemic period [25, 26]. Therefore, ρ_c is estimated but is interpreted only as an effective observation parameter, not as a direct estimate of any single reporting mechanism. The dispersion parameter $\phi_c > 0$ accounts for residual overdispersion in weekly counts. Larger ϕ_c corresponds to weaker overdispersion, with the Poisson model recovered as $\phi_c \to \infty$.

1.4 Fixed initial conditions and estimated parameters

Initial conditions and spin-up. Let t_{pre} denote the first day of epidemiological week 201001. For each city c, the initial infectious human stock is fixed as

$$I_{h,c}^0 = \max\{1, Y_{c,w_{\text{pre}}}\}, \tag{23}$$

where $Y_{c,w_{\text{pre}}}$ is the probable-case count in the first available week. The remaining human compartments are initialized by

$$E_{h,c}(t_{\text{pre}}) = R_{h,c}(t_{\text{pre}}) = 0, \qquad S_{h,c}(t_{\text{pre}}) = N_{h,c} - I_{h,c}^0. \tag{24}$$

This is a study-specific simplifying convention, not evidence that no latent infection or previous immunity was present. Simplified initializations and pre-simulation periods are commonly used to reduce the influence of uncertain initial states in dengue models [27, 3].

The adult female mosquito population is initialized as

$$S_{m,c}(t_{\text{pre}}) = 0.7N_{h,c}, \qquad E_{m,c}(t_{\text{pre}}) = I_{m,c}(t_{\text{pre}}) = 0. \tag{25}$$

The factor 0.7 is adapted from a weather-driven dengue framework applied to Brazilian municipalities [28]. Because vertical transmission is excluded, no infected mosquitoes are introduced initially. All initial conditions are fixed rather than estimated. The full system is then propagated during the complete 2010 spin-up year, and observations from that year do not enter the fitting objective.

Estimated parameters and constraints. The trainable quantities are:

- the shared neural parameters $\boldsymbol{\theta}$ for the four development cities;
- the city-specific recruitment scales κ_c;
- the city-specific effective observation probabilities ρ_c;
- the city-specific negative-binomial dispersion parameters ϕ_c.

All biological thermal-response parameters, human progression and recovery rates, human mortality, population sizes, and initial conditions are fixed.

Constraints are imposed by unconstrained working parameters:

$$\kappa_c = \operatorname{softplus}(\widetilde{\kappa}_c), \qquad \rho_c = \sigma(\widetilde{\rho}_c), \qquad \phi_c = \operatorname{softplus}(\widetilde{\phi}_c), \tag{26}$$

where $\operatorname{softplus}(u) = \log(1 + \exp u)$. This guarantees $\kappa_c > 0$, $0 < \rho_c < 1$, and $\phi_c > 0$ throughout optimization.

For C development cities, the training objective is the equally weighted mean negative log-likelihood plus neural weight and temporal-smoothness penalties:

$$\begin{aligned} \mathcal{L} = {} & \frac{1}{C}\sum_{c=1}^{C}\frac{1}{n_c}\sum_{w\in\mathcal{W}_{c,\mathrm{train}}} -\log p_{\mathrm{NB}}\left(Y_{c,w} \mid \rho_c C_{c,w}, \phi_c\right) \\ & + \lambda_\theta \|\boldsymbol{\theta}\|_2^2 + \lambda_s \frac{1}{C}\sum_{c=1}^{C}\frac{1}{n_c - 1}\sum_{w}\left[q_{\boldsymbol{\theta}}(\boldsymbol{z}_{c,w}) - q_{\boldsymbol{\theta}}(\boldsymbol{z}_{c,w-1})\right]^2. \end{aligned} \tag{27}$$

Equal city weighting is a study-specific design choice that prevents the largest municipality from determining the shared network solely through its case-count scale. The L_2 penalty limits unnecessarily large network weights, while the smoothness term discourages week-to-week oscillation in a latent recruitment response driven by slowly varying lagged climate. Their values are fixed or selected exclusively on the validation period, as described in Section 2.5.

2 Data and numerical implementation

2.1 Data sources and study cities

The analysis considers Belém, Salvador, Rio de Janeiro, São Paulo, and Brasília. Weekly probable dengue cases are obtained from SINAN and IBGE records organized by InfoDengue. Cases are indexed by municipality and epidemiological week of symptom onset, with each week beginning on Sunday [11, 22].

Meteorological data are derived from ERA5 reanalysis and organized by the Mosqlimate project [29, 30]. The primary covariates are weekly mean temperature, weekly accumulated precipitation, and weekly mean relative humidity. Demographic data are obtained from the InfoDengue population table, which contains annual municipality populations from 2010 onward [11]. To remain consistent with the constant-population assumption, the 2010 municipality population is used as $N_{h,c}$ throughout the simulation.

All epidemiological, meteorological, and demographic records are matched through the official IBGE municipality geocode. The five-city model-ready panel is restricted to epidemiological weeks 201001–202252. Thus, all model development, descriptive figures, calibration, validation, and testing stop in 2022. Data from 2023–2024 are not used or displayed in the present analysis.

Table 2: Study cities, municipality identifiers, and modelling roles.

City	IBGE geocode	State	Role	Shared-network learning
Belém	1501402	PA	Development city	Included
Salvador	2927408	BA	Development city	Included
Rio de Janeiro	3304557	RJ	Development city	Included
São Paulo	3550308	SP	Development city	Included
Brasília	5300108	DF	Spatially held-out city	Excluded

Brasília is excluded from shared-network learning because it represents an inland climate with a pronounced wet–dry seasonal contrast relative to the development set [31]. The evaluation is therefore a *spatial transfer of the shared climate–recruitment function with local calibration*, followed by a temporal out-of-sample test. It is not a zero-calibration transfer: the shared neural parameters remain frozen, but Brasília-specific κ_c, ρ_c, and ϕ_c are estimated from Brasília data before the 2021–2022 test.

2.2 Data preprocessing and quality control

The final panel contains one row for each city and epidemiological week from 3 January 2010 to 25 December 2022, corresponding to 678 weeks per city. The following preprocessing steps are applied in the stated order.

1. **Selection and harmonization.** The five municipality geocodes are selected, dates are converted to the first day of the epidemiological week, and records are sorted by city and date. Epidemiological, climate, and demographic tables are merged using $(\text{geocode}, \text{epiweek})$.

2. **Uniqueness and weekly continuity.** Duplicate municipality–week rows are collapsed before merging, and a complete Sunday-based weekly calendar is constructed for every city. The harmonized model-ready panel contains no duplicated or missing municipality–week records over 2010–2022; therefore, no statistical imputation is applied in the primary analysis.

3. **Epidemiological outcome.** The variable $Y_{c,w}$ is the non-negative integer count of probable dengue cases in week w, defined as suspected cases minus discarded cases in the source documentation [11]. Counts are retained on their original scale for the negative-binomial observation model; population-standardized incidence may be shown only in supplementary descriptive figures.

4. **Temperature and relative humidity.** The variables $T_{c,w}$ and $H_{c,w}$ are the weekly mean temperature in degrees Celsius and weekly mean relative humidity in percent, respectively. Values are checked for finite values and physically admissible ranges before lag construction.

5. **Weekly accumulated precipitation.** The released climate documentation states that hourly data were first summarized by day and that daily measures were then averaged within each epidemiological week [11]. Consequently, if $\overline{p}_{c,w}$ denotes the released weekly mean of daily total precipitation, weekly accumulation is reconstructed as

$$P_{c,w} = 7\,\overline{p}_{c,w}. \tag{28}$$

   This conversion is applied before the logarithmic transformation in Equation (9).

6. **Training-only standardization.** $P_{c,w}$ is transformed to $P^{\star}_{c,w}$, and pooled means and standard deviations are calculated from the four development cities over 2011–2018 only. The same fixed statistics are then used for every later period and for Brasília.

7. **Lag construction.** Lagged vectors are constructed separately within each city after sorting by week. The 2010 spin-up supplies the climate history required at the beginning of 2011, so no scored training week is removed because of missing lags for any candidate $L \leq 7$.

The use of reanalysis-based meteorological data produces a spatially and temporally complete forcing series, but it does not remove uncertainty associated with gridded climate products or their representation of within-city conditions [29].

2.3 Spatial and temporal data partition

The year 2010 is used as a common spin-up period. The model is propagated under observed meteorological forcing, but case observations from that year do not contribute to the fitting objective. Pre-simulation reduces the influence of uncertain initial states on evaluated trajectories [3].

For the four development cities, 2011–2018 data are used to estimate the shared network and city-specific parameters. Data from 2019–2020 are used only for architecture and regularization selection. The chronological order is preserved to prevent future observations from entering model development [32, 15, 33].

After hyperparameter selection, the shared component is refitted on the complete 2011–2020 development data and then frozen. In Brasília, data from 2011–2020 are used only to estimate the local parameters κ_{Brasilia}, ρ_{Brasilia}, and ϕ_{Brasilia}. The final external test is conducted over 2021–2022 without further parameter or hyperparameter updating.

Table 3: Spatial and temporal partition of the study data.

Cities	**Period**	**Stage**	**Purpose**
All five cities	2010	Spin-up	Dynamic initialization; cases not scored
Four development cities	2011–2018	Training	Shared and local parameter estimation
Four development cities	2019–2020	Validation	Hyperparameter selection and early stopping
Four development cities	2011–2020	Final refitting	Final shared neural component
Brasília	2011–2020	Local calibration	Local parameters with shared network frozen
Brasília	2021–2022	External test	Final spatial-transfer and temporal test

2.4 Numerical integration

The coupled system is implemented with differentiable tensor operations in Python using PyTorch. To keep the numerical procedure transparent and fully compatible with gradient-based training, the ODE is integrated with the classical fourth-order Runge–Kutta method using a fixed step $\Delta t = 1$ day [34]. The epidemiological and climate forcings are constant within each epidemiological week and are therefore applied over seven consecutive daily integration steps. The weekly latent incidence in Equation (20) is accumulated with the same Runge–Kutta stages.

The mathematical system preserves non-negative states. Numerically, values within floating-point tolerance below zero are set to zero after a step; larger negative values cause the optimization run to be rejected. As a numerical verification, the final fitted trajectories are recomputed using $\Delta t = 0.5$ day. A relative difference below 1% in weekly fitted means and evaluation metrics is used as a study-specific convergence check.

2.5 Training and hyperparameter selection

The neural network and city-specific parameters are estimated jointly by automatic differentiation, following the differentiable-programming strategy used for universal differential equations [2, 3]. Network weights are initialized with Glorot initialization and biases are initialized at zero [35]. The local parameter initializations are

$$\rho_c^{(0)} = 0.10, \qquad \phi_c^{(0)} = 10, \qquad \kappa_c^{(0)} = 1.4\,\overline{\mu_m(T)}_{c,\text{train}}, \tag{29}$$

where the last expression approximately balances mortality of the initial mosquito population $0.7N_h$ when the initial sigmoid output is near 0.5. The value $\rho_c^{(0)} = 0.10$ is only an optimization starting value and is not fixed or used as an informative prior; it reflects the substantial under-ascertainment documented for passive dengue surveillance [25, 26].

Optimization uses Adam with learning rate 10^{-3} [36]. Each candidate model is trained for at most 2000 epochs, with early stopping when validation negative log-likelihood has not improved for 100 consecutive epochs. Three random initializations are used for each candidate configuration.

The deliberately small candidate set is

$$L \in \{3, 5, 7\}, \qquad J \in \{4, 8\}, \qquad \lambda_s \in \{0, 10^{-3}\}, \qquad \lambda_\theta = 10^{-4}. \tag{30}$$

The configuration and random initialization with the lowest mean validation negative log-likelihood over the four development cities is selected. The epoch corresponding to the best validation value is recorded. The selected configuration is then refitted on 2011–2020 development data for that fixed number of epochs, avoiding the use of 2021–2022 test information.

After the shared parameters $\boldsymbol{\theta}$ are frozen, Brasília-specific κ_c, ρ_c, and ϕ_c are calibrated on 2011–2020 using Adam for at most 1000 epochs. Local calibration stops if the relative improvement in its objective is smaller than 10^{-6} for 100 consecutive epochs. Random seeds, software versions, and the selected hyperparameters are stored with the analysis outputs to support reproducibility [33].

2.6 Comparison models and ablation analyses

The full hybrid model is compared with three prespecified alternatives. All alternatives retain the same SEI–SEIR equations, observation model, data partition, optimizer, and evaluation metrics, and all their free parameters are re-estimated.

1. **Constant-recruitment model.** The neural term is removed and $\Lambda_{m,c}(t) = \kappa_c N_{h,c}$ is constant. This baseline tests whether a learned climate-sensitive recruitment function improves fit and transfer relative to a climate-independent recruitment pressure.

2. **No-memory hybrid model.** The network receives only the current-week variables $(T_w, P^\star_w, H_w)$, corresponding to $L = 1$. This tests the contribution of lagged climate.

3. **No-humidity ablation.** The network receives lagged temperature and precipitation but excludes relative humidity. This tests whether humidity contributes information beyond temperature and precipitation.

These comparisons are kept deliberately limited so that each one addresses a specific modelling question: the value of the neural climate component, the value of meteorological memory, and the additional value of relative humidity. Prespecifying comparison models before examining the test results is consistent with transparent epidemic-model evaluation [33].

2.7 Evaluation metrics and predictive uncertainty

Hyperparameters are selected using the mean negative-binomial negative log-likelihood on 2019–2020 validation data. The negative log-likelihood is a proper probabilistic score when the complete predictive distribution is specified [37]. Final performance is reported separately for each stage and especially for the untouched Brasília 2021–2022 test period.

Let $\widehat{m}_{c,w}$ denote the fitted or predicted weekly mean. The point-prediction metrics are

$$\mathrm{MAE}_c = \frac{1}{n_c} \sum_{w=1}^{n_c} \left| Y_{c,w} - \widehat{m}_{c,w} \right|, \tag{31}$$

$$\mathrm{RMSE}_c = \sqrt{\frac{1}{n_c} \sum_{w=1}^{n_c} \left(Y_{c,w} - \widehat{m}_{c,w} \right)^2}. \tag{32}$$

MAE and RMSE are reported together because they summarize absolute error with different sensitivity to large misses [38]. Percentage errors are not used because weekly dengue series contain zero counts, for which percentage measures may be undefined or unstable [38].

Epidemic timing is summarized by the absolute peak-week error

$$E_{\mathrm{peak},c} = \left|\arg\max_w Y_{c,w} - \arg\max_w \widehat{m}_{c,w}\right|, \tag{33}$$

expressed in epidemiological weeks.

For each week, the 2.5% and 97.5% quantiles of the fitted negative binomial distribution define a central 95% predictive interval $[\ell_{c,w}, u_{c,w}]$. Interval calibration and sharpness are summarized by

$$\mathrm{Coverage}_{0.95,c} = \frac{1}{n_c}\sum_{w=1}^{n_c} \mathbf{1}\left\{\ell_{c,w} \le Y_{c,w} \le u_{c,w}\right\}, \tag{34}$$

$$\mathrm{Width}_{0.95,c} = \frac{1}{n_c}\sum_{w=1}^{n_c} \left(u_{c,w} - \ell_{c,w}\right). \tag{35}$$

Coverage and interval width are interpreted jointly because well-calibrated intervals should include observations at approximately their nominal rate without being unnecessarily wide [39, 37].

The primary predictive intervals represent observation uncertainty implied by the negative-binomial model. They do not constitute full Bayesian parameter uncertainty intervals. Optimization stability is assessed separately by the three random initializations: variation in validation/test metrics and fitted local parameters is reported in supplementary material. Parameters showing large between-start variability are not given a strong mechanistic interpretation, consistent with known practical-identifiability concerns in vector-borne disease models [17].

2.8 Exploratory visualization of the modelling data

Figure 2 presents weekly dengue notifications for the five cities over 2010–2022. The dashed horizontal line represents the city-specific historical weekly median, while the labelled point identifies the largest observed weekly peak. Brasília is displayed as the spatially held-out city and is excluded from shared-network development.

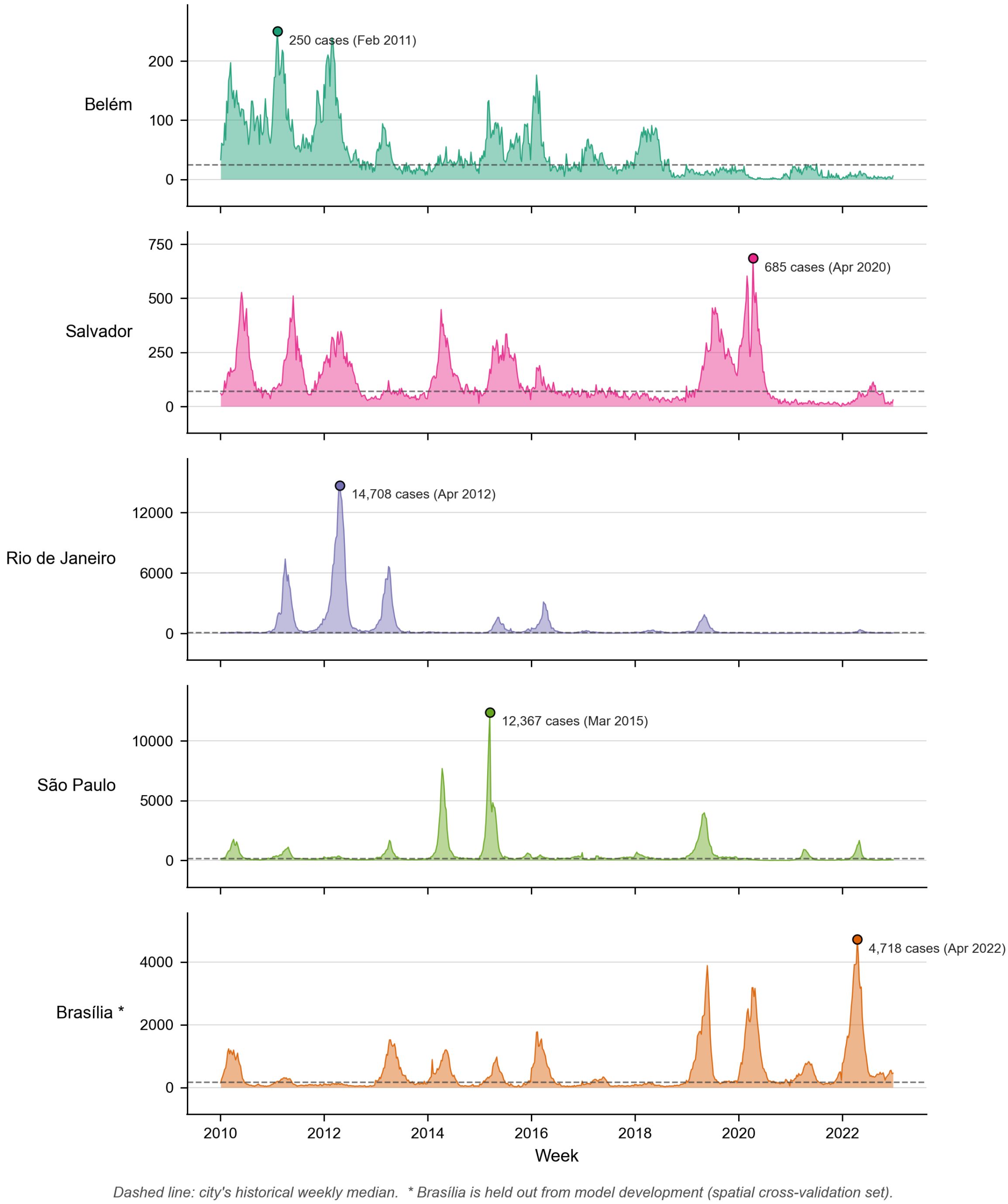

Figure 2: Weekly probable dengue cases in Belém, Salvador, Rio de Janeiro, São Paulo, and Brasília from 2010 to 2022. The dashed horizontal line represents the city-specific historical weekly median. The labelled point indicates the maximum weekly number of probable cases. Brasília is excluded from shared-network learning.

Figure 3 summarizes the seasonal climatology of weekly meteorological covariates over 2010–2022. Monthly values correspond to the mean weekly temperature, accumulated precipitation, and relative humidity calculated across the study period, with shaded bands representing 95% confidence intervals.

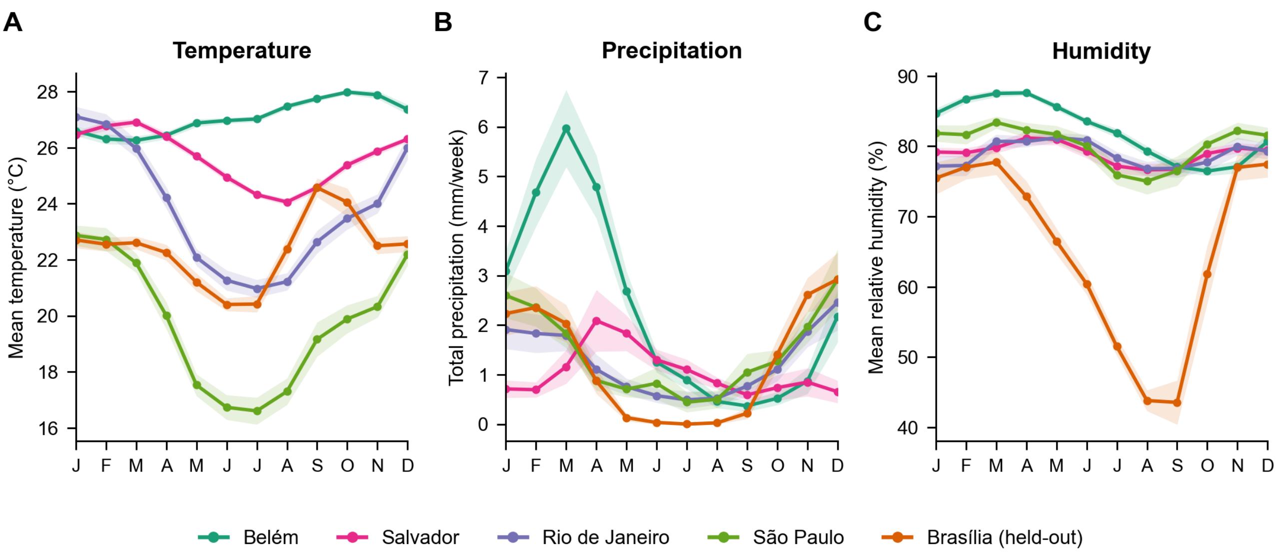

Figure 3: Seasonal climatology of the meteorological covariates by city over 2010–2022: **(A)** weekly mean temperature, **(B)** weekly accumulated precipitation, and **(C)** weekly mean relative humidity. Lines represent monthly means, and shaded areas represent 95% confidence intervals.

The complete distributions of weekly meteorological observations are shown in Figure 4. Violin plots represent empirical distributions, while embedded boxplots indicate the median and interquartile range.

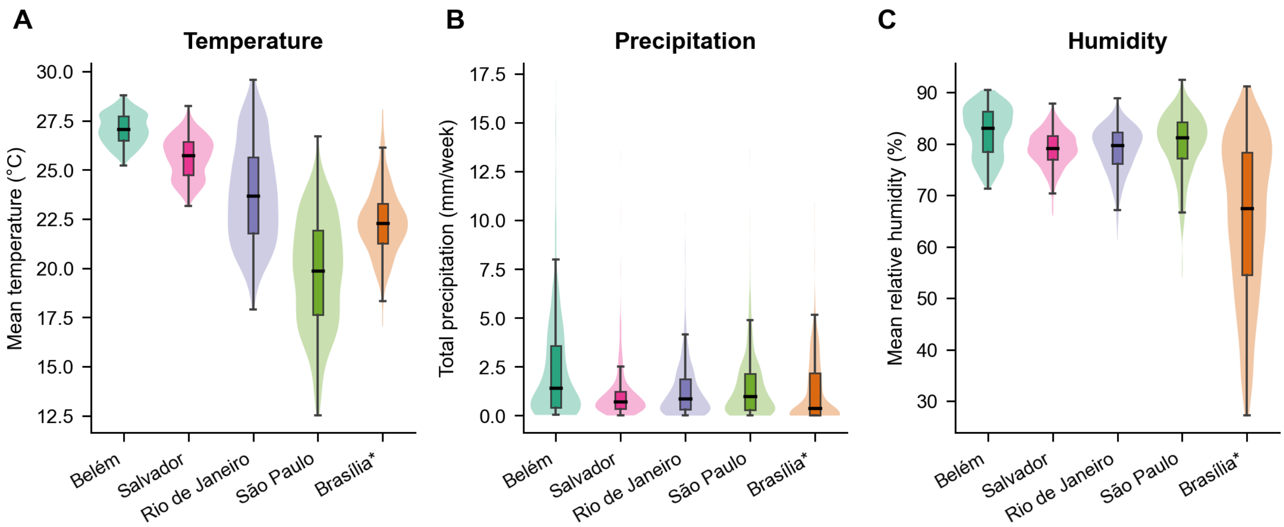

Figure 4: Distribution of weekly meteorological variables by city during 2010–2022: **(A)** mean temperature, **(B)** accumulated precipitation, and **(C)** mean relative humidity. Violin plots represent empirical densities, and embedded boxplots summarize the median and interquartile range. The asterisk identifies Brasília as the spatially held-out city.

The exploratory figures show substantial spatial and temporal heterogeneity. Before 2023, the largest weekly peaks occur at different times across cities: 2011 in Belém, 2012 in Rio de Janeiro, 2015 in São Paulo, 2020 in Salvador, and 2022 in Brasília. Climatically, Belém remains warm and humid throughout the year, São Paulo exhibits stronger annual temperature variation, and Brasília shows a pronounced dry-season reduction in precipitation and relative humidity. These contrasts motivate city-specific epidemiological scaling together with a shared climate-sensitive recruitment function; however, visual associations alone do not establish a causal relationship between climate and dengue incidence.

A Mathematical analysis

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